Accident in a Co-generation Plant

Analysis of the Human and Organisational Factors (HOF) involved

Minor Assignment | Technology Risk Governance

MSc in Cyber Risk Strategy and Governance | Prof. Paolo Trucco

Team members

Name Surname (ID) Name Surname (ID) Name Surname (ID)

Events 1 and 3 (completed and summarised from the class discussion) and two further events of the time-line (own analysis).

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Approach and reading key

Method

Each event of the accident time-line is analysed with the **behavioural approach** (Reason's model of organisational accidents), starting from the *phenotype* (how the event manifests) and going back to the *genotypes* (local conditions and latent factors). For every event we report:

- **one page** with the template compiled in bullet form, following the scheme used in class for Events 1 and 3: human factors pathway on the left, system factors pathway on the right, organisational latent errors on top;
- **one page** of justification, explaining which information of the case text supports each box, the classification of the active error and a short **cognitive view (CREAM)** of the human contribution.

Events 1 and 3 reproduce and complete the solution discussed in class; the other events are our own analysis.

How to read the templates

- **Event**: the event occurs if at least one of the two is present: a failure of a technical defence, barrier or safeguard, or an active human error.
- Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it, with the technical latent error behind it). **White boxes**: contributing conditions.
- **Top box vs. side boxes** (scope of influence): a latent factor that affects only one pathway goes into that pathway's box; a factor that affects both the human and the technical side goes on top.

Sources

[§x.y] section of the case document; [Fig. 3] event time-line; [Class] discussion of the case in class; [Inf.] our inference, not stated explicitly in the case.

Event time-line and overview

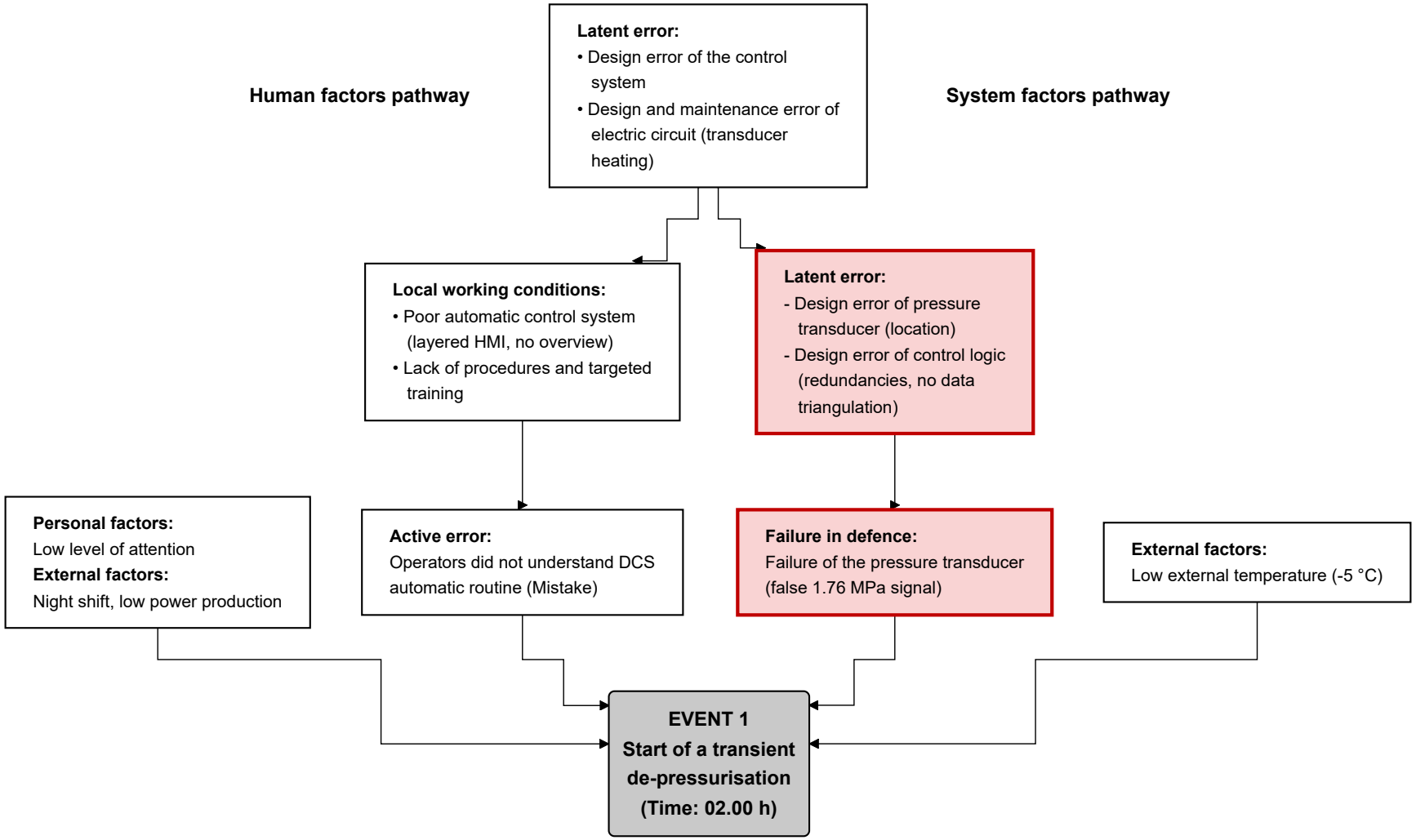
#	Time	Event	Dominant pathway
1	02.00	Pressure transducer failure, start of de-pressurisation	System
2	02.04	Manual activation of main circulation pumps	Human + system
3	02.41	Opening of relief valves, overflow of storage tank, flooding of pumping room	Human + system
4	03.00	Manual closure of relief valves (very late)	Human + system
5	03.20	Fire protection intervention, remote control out of service	System
6	03.45	Temperature increase at heat-exchanger outlet	Human + system
7	04.39	Circulation pumps started by-passing low-pressure protection	Human (violation) + system
8	05.15	Manual start-up of heat recovery from turbo-gas	Human + system
9	06.15	Shock wave (water hammer)	Final event

Emergency phases (causal and contributing factors)

- **From the control room**: 1st de-pressurisation and vaporisation (Events 1, 2, with 4).
- **On the field**, after the loss of remote control: 2nd de-pressurisation (3, 5), vaporisation due to heat exchange (4, 6, 8) and peak pressurisation (7), all converging on the shock wave (9).

Event 1 | Start of a transient de-pressurisation (02.00 h)

The outdoor pressure transducer freezes during a cold winter night and sends a false reading of 1.76 MPa while the real pressure is normal. The automatic control system reacts to the false signal: pressurising pumps slowed down, relief and expansion-tank valves opened, main feed pumps stopped. A rapid de-pressurisation starts.



Active error: Omission (Swain & Guttman) | Mistake (Reason) | Knowledge-based (SRK) | CREAM: Timing → Omission. **Dominant pathway**: system (technical failure of the pressure measurement).

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Template reproduces the class solution; items in parentheses complete it with the class discussion.

Event 1 | Justification of the template

Evidence from the case, box by box

Failure in defence (red). The transducer, installed outdoors near an expansion tank, froze together with its signal piping in a cold but "not exceptional" night and sent 1.76 MPa while the real pressure was normal [§2, ev. 1]. **External factors:** low external temperature, about -5 °C [Class].

Latent error, system (red). A safety-critical measurement was placed outdoors, protected only by an electric heating circuit (*location*) [§2; Class]. One single transducer drove both pump speed and pressure valves [§2]: no redundant source and no *data triangulation*, although 1.76 MPa was thermodynamically incompatible with the measured temperature and flow [Class]. The heater failure was also invisible, because the monitoring of the heaters' power supply was wrongly designed [§3.3].

The DCS then did exactly what it was designed for, but on wrong data: it acted as a barrier against a non-existent over-pressure. This is a loss of data *integrity*: the same scenario could be triggered by a cyber attack on the sensor signal [Class].

Active error (white). Operators did not understand what the automatic routine was doing and lost the chance to build situational awareness (*mistake*). Evidence comes from the phenotype of Event 2: their first action was to restart the circulation pumps [Class]. The box is white because the error did not create the de-pressurisation; it removed an early recovery opportunity.

Local working conditions. HMI made of layered windows, without an overview, and no warning for an "abnormal measure" out of the transducer range [§3.3]; no emergency procedures [§3.4]; training only on the job [§3.7].

Personal and external factors. Night shift with base-load demand: almost no interaction between operators and system, hence low attention and low readiness when the transient started [§3.6; Class].

Latent error, organisational (top). Design error of the control system and design & maintenance error of the electric circuit influence *both* pathways: they make the barrier fail and leave operators unable to understand it. Maintenance is also organisational: not providing procedures and training means not maintaining the operators' situational awareness as a defence [Class].

Classification of the active error

Phenotype (Swain & Guttman)	Omission: no diagnosis of, and no intervention on, the automatic routine.
Reason's taxonomy	Mistake: wrong understanding of the situation at the intention stage.
SRK level (Rasmussen)	Knowledge-based (diagnose a DCS acting on false data); the rule-based support was missing (no procedure for this transient).

Cognitive view (CREAM, Hollnagel 1998)

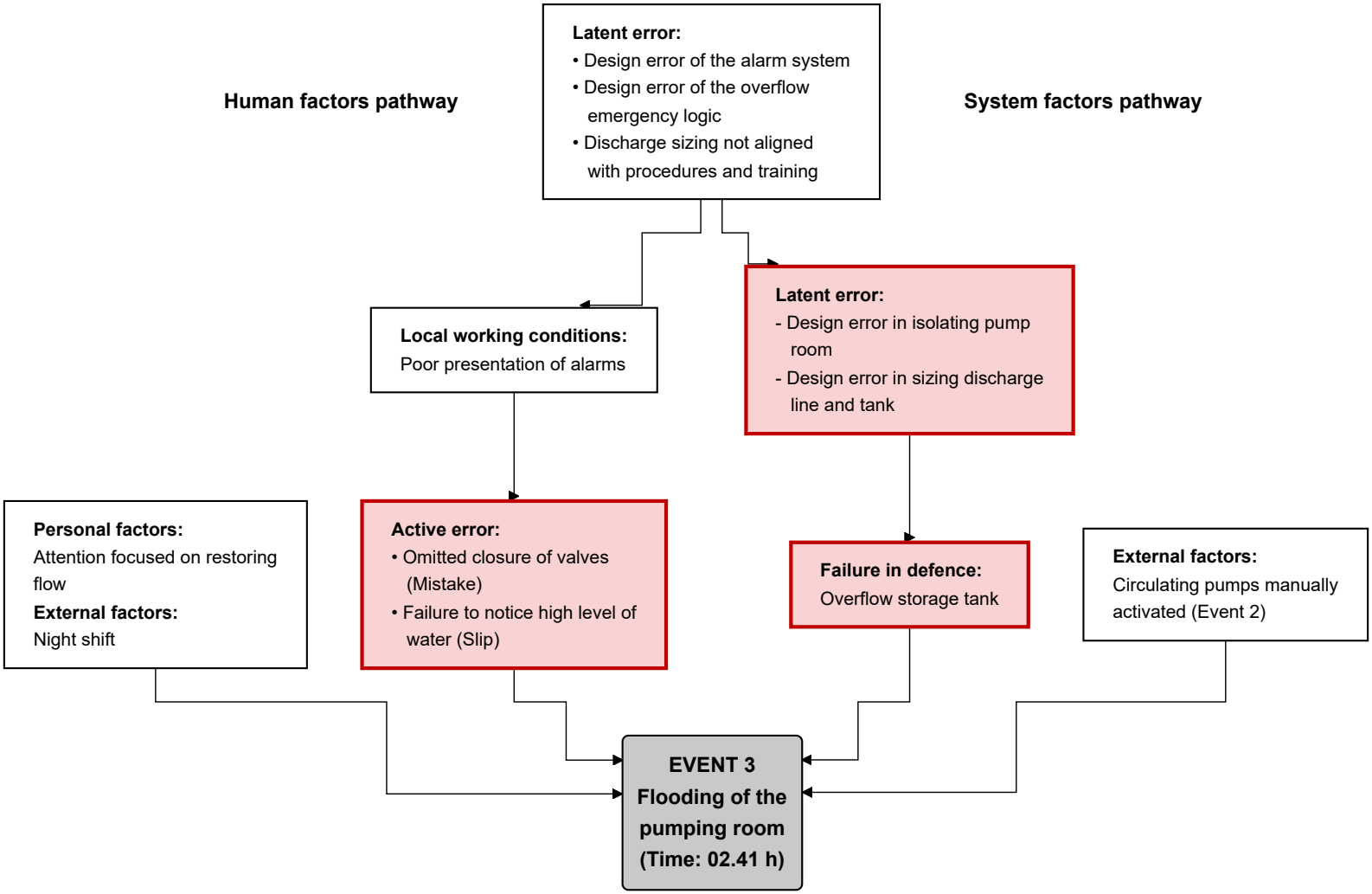
Socio-technical context	Organisation: no emergency procedures, shift team undersized for emergencies. Technology: layered HMI, no out-of-range alarm. Environment: night shift, stable base load.
Cognitive function	Observation → Interpretation (the automatic actions were not noticed or not interpreted).
Individual factors	Psychological aspects (low attention at night); experience and skill (on-the-job training only).
Error mode	Timing → Omission.
General antecedents	Inattention; Observation missed; Faulty diagnosis; Inadequate procedure.
Specific antecedent	None (first event of the sequence).

Improvement levers

- **Prevention**: relocated and redundant pressure sensors with plausibility checks (pressure vs. temperature and flow); alarm on heater supply failure.
- **Recovery**: out-of-range ("abnormal measure") alarm and an overview display of the automatic actions in progress.
- **Emergency management**: procedure and training for the de-pressurisation transient; integrity monitoring of OT sensor data (cyber).
- After the accident the company relocated and duplicated the sensors and replaced most of the DCS [Class].

Event 3 | Flooding of the pumping room (02.41 h)

The storage tank reaches its maximum level and water is discharged through the relief valves. The emergency discharge system is not dimensioned for this flow and boiling water starts to flood the pumping room.



Active errors: Omission (Swain & Guttman) | Mistake + Slip (Reason) | Skill-based (SRK) | CREAM: Timing → Omission. **Dominant pathway**: human and system together (alignment between technical and organisational design).

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Template reproduces the class solution; the personal/external factors box and the third top item complete it with the class discussion.

Event 3 | Justification of the template

Evidence from the case, box by box

Event and external factors (system). The maximum level in the storage tank resulted in water discharge through the relief valves; the emergency system was not dimensioned for the flow and boiling water flooded the pumping room [§2, ev. 3]. The circulation pumps manually activated in Event 2 pushed out a flow 4–5 times higher than any value considered by the designers [Class].

Failure in defence (red). Overflow of the storage tank: the last barrier collecting the discharged water was exceeded [§2, ev. 3].

Latent error, system (red). Discharge line and tank were sized for the normal steam condensate released by safety valves, not for the full flow of the pumps; the pumping room was not isolated from the discharge [Class].

Active errors (red). (1) *Omitted closure of the valves* (mistake): with relief valves open, checking whether the pressure is really high and closing the valves before restarting the pumps is a basic operation. (2) *Failure to notice the high water level* (slip): the alarm was available but not perceived [Class].

Local working conditions. The "high level of water in the storage tank" alarm was shown, but dispersed in a long list of less important alarms; HMI made of layered windows with no general overview [§3.3]; no video surveillance of the rooms [§3.3].

Personal and external factors (completion). Night shift [§3.6]; attention focused on restoring the flow, as shown by Event 2 [Class].

Why no "lack of training" here. Both errors are at the skill-based level: basic competences acquired with entry-level training and experience, which must be supported by a clear alarm presentation. Training is decisive for Event 1 (rule- and knowledge-based); it may be added here only if specified as entry-level training [Class].

Latent error, organisational (top). Design errors of the alarm system and of the overflow emergency logic. The size of the tank and the operators' procedures and training were decided separately: a tank is large enough only if operators can detect and close the valves *before* it fills. This misalignment between technical and organisational design affects both pathways, so it goes on top [Class].

Classification of the active error

Phenotype (Swain & Guttman)	Omission: required action not performed (valve closure); omission of detection (alarm not perceived).
Reason's taxonomy	Mistake (omitted closure, wrong intention) + Slip (alarm missed at execution/perception level).
SRK level (Rasmussen)	Skill-based: elementary routine operations, like changing gear while driving.

Cognitive view (CREAM, Hollnagel 1998)

Socio-technical context	Organisation: no procedures, team undersized for emergencies, tank sizing not aligned with operator response time. Technology: alarm flooding, layered HMI, no CCTV. Environment: night shift.
Cognitive function	Observation (high-level alarm missed) → Planning (closure not planned).
Individual factors	Psychological aspects (attention narrowed on flow restoration).
Error mode	Timing → Omission.
General antecedents	Observation missed; Inattention.
Specific antecedent	Earlier omission (no diagnosis of the de-pressurisation in Events 1–2).

Improvement levers

- **Prevention**: size discharge line and tank jointly with procedures, comparing time-to-overflow with time-to-detect-and-act; isolate the pumping room.
- **Recovery**: alarm prioritisation (no long undifferentiated lists), overview display, CCTV of critical rooms.
- **Emergency management**: entry-level skills checked periodically, not only on the job.